

1 Irrigation strategies under contrasting water-storage and rainfall

2 conditions in the North China Plain

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10 Abstract

11 Efficient irrigation allocation is essential for sustaining grain production while limiting water consumption in the
12 North China Plain wheat–maize rotation. How spatial targeting redistributes production losses, and whether
13 regional water-storage information improves rainfall-based irrigation decisions, remain unresolved. Continuous
14 process-based crop simulations compared uniform and targeted allocations under equal irrigation budgets and
15 evaluated annual storage–rainfall strategies against matched rainfall-only controls during 2014–2025. Growth
16 parameters were fitted to multisite observations; water-use parameters were calibrated against Wuqiao treatments in
17 2016–2018 and retrospectively tested in 2019. At a 50% irrigation reduction, targeting retained 0.73 t ha⁻¹ yr⁻¹
18 more simulated combined dry grain than uniform cuts across 9.87 million rotation hectares, equivalent to 7.22
19 million t yr⁻¹ or 4.87% of uniform production. Mean production was lower than under uniform cuts on 37.60% of
20 mapped area, while crop-plus-fallow evapotranspiration increased by 20.76 mm yr⁻¹ (2.05 km³ yr⁻¹). Across nine
21 years with eligible Gravity Recovery and Climate Experiment (GRACE) observations, strategies selected at 95%
22 and 98% grain targets retained 96.27% and 99.63% of conventional production, respectively. The 95% strategy
23 reduced irrigation by 130.72 mm and evapotranspiration by 36.84 mm per rotation. Matched storage–rainfall and
24 rainfall-only strategies selected identical quotas and produced identical outcomes. These conditional results support
25 joint production and consumptive-use constraints and local production safeguards, while showing that additional
26 storage information provides no decision benefit within the tested quota design.

27 **Keywords**

28 winter wheat–summer maize rotation; irrigation management; crop modeling; GRACE; water availability;
29 consumptive water use

30 **1. Introduction**

31 Irrigation management in the North China Plain must reconcile grain production with water-resource constraints
32 across the winter wheat–summer maize rotation (Yang et al., 2022). Wheat grows through a relatively dry season,
33 whereas maize receives more summer rainfall; soil-water reserves connect their responses to irrigation (Yang et al.,
34 2024). Long-term well-watered observations show that rising grain yield has accompanied increasing actual
35 evapotranspiration (ET), despite comparatively stable reference ET (Zhang et al., 2011). Maintaining production
36 with less irrigation therefore requires attention to both the water delivered to fields and the water consumed over the
37 complete rotation.

38 The timing and location of irrigation supply determine its marginal production return. A sixteen-year experiment in
39 2003–2018 compared seven wheat–maize irrigation schedules and found diminishing production gains as irrigation
40 increased (Yang et al., 2022). A four-year Wuqiao experiment also showed that precipitation conditions influenced
41 the preferred wheat schedule and grain production in the subsequent maize crop (Yang et al., 2024). These
42 responses reflect the sequence of crop demand, rainfall and soil-water carry-over across both seasons (Yang et al.,
43 2024; Yang et al., 2022). At regional scale, irrigation requirements vary among locations and hydrological years
44 (Ren et al., 2022; Wu et al., 2019). Crop-model optimization and irrigation-allocation studies provide approaches
45 for using this heterogeneity to guide management (Jiang et al., 2016; Xiao et al., 2024). Under a common water
46 budget, targeting locations with larger marginal grain returns can improve regional production retention, while
47 changing the distribution of production losses.

48 Assessing these allocation consequences requires consistent crop and water accounting. Field delivery, ET, return
49 flows and storage changes describe different quantities, so a reduction in irrigation does not directly establish an
50 equivalent reduction in consumption (Grafton et al., 2018; Perry et al., 2009). Process-based applications of APSIM
51 and RZ-SHAW in the North China Plain connect crop production with water productivity, field water balance and
52 ET (Chen et al., 2010; Fang et al., 2014). However, transpiration formulation remains a major source of variation in
53 simulated wheat water use (Cammarano et al., 2016). Phenology, canopy, biomass, grain and ET observations

54 therefore constrain complementary processes, while documented irrigation treatments assess their combined
55 response to water supply. Continuous simulation retains the soil-water carry-over needed to compare complete
56 rotation histories.

57 Spatial allocation determines where irrigation is supplied, while antecedent hydrological conditions can inform the
58 annual irrigation budget. Recent rainfall describes water input at the crop-system scale, whereas GRACE total water
59 storage integrates several terrestrial stores over a regional footprint (Save et al., 2016; Scanlon et al., 2016). Recent
60 aquifer recovery under water transfers, abstraction management and changing climate illustrates the broader context
61 of regional storage variation (Long et al., 2025; Xu et al., 2026). Storage information may thus characterize
62 conditions beyond local rainfall. Its practical value for irrigation nevertheless depends on whether it changes the
63 selected action or improves the resulting grain–water outcome relative to rainfall-based decisions.

64 Two connected management questions remain unresolved. First, it is unclear how much grain spatial targeting can
65 retain under an identical irrigation budget, and what consumptive-water costs and persistent local production losses
66 accompany that regional benefit. Rotation-wide accounting must retain soil-water carry-over to assess these
67 outcomes together. Second, the additional decision value of regional storage information beyond antecedent rainfall
68 remains uncertain. A storage indicator can distinguish hydrological conditions without changing an irrigation
69 choice. Comparisons using the same production targets, candidate quotas and monitoring availability are needed to
70 isolate its contribution to management outcomes.

71 The objectives were to (i) quantify rotation grain production, crop-plus-fallow ET and profile drainage under
72 uniform and spatially targeted irrigation allocations with common regional water budgets; (ii) characterize the
73 spatial distribution, magnitude and annual persistence of allocation-related grain gains and losses; and (iii) assess
74 whether antecedent regional storage information changes annual irrigation quotas and grain–water outcomes
75 relative to rainfall-only controls under matched selection and monitoring conditions. Continuous Open Crop Model
76 simulations of the North China Plain wheat–maize rotation, parameterized using multisite crop observations and
77 documented irrigation treatments, provide the basis for these conditional comparisons during 2014–2025.

2. Materials and methods

2.1. Overall workflow

The workflow links crop-model calibration and evaluation to regional irrigation-management comparisons through four steps (Figure 1). First, crop development and growth are calibrated and evaluated, and crop water-use parameters are fitted to irrigation-treatment responses and tested retrospectively. The resulting parameters remain fixed for the management comparisons. Second, continuous winter wheat–summer maize simulations quantify crop responses to five irrigation levels, with soil water carried through successive crops and intervening fallows.

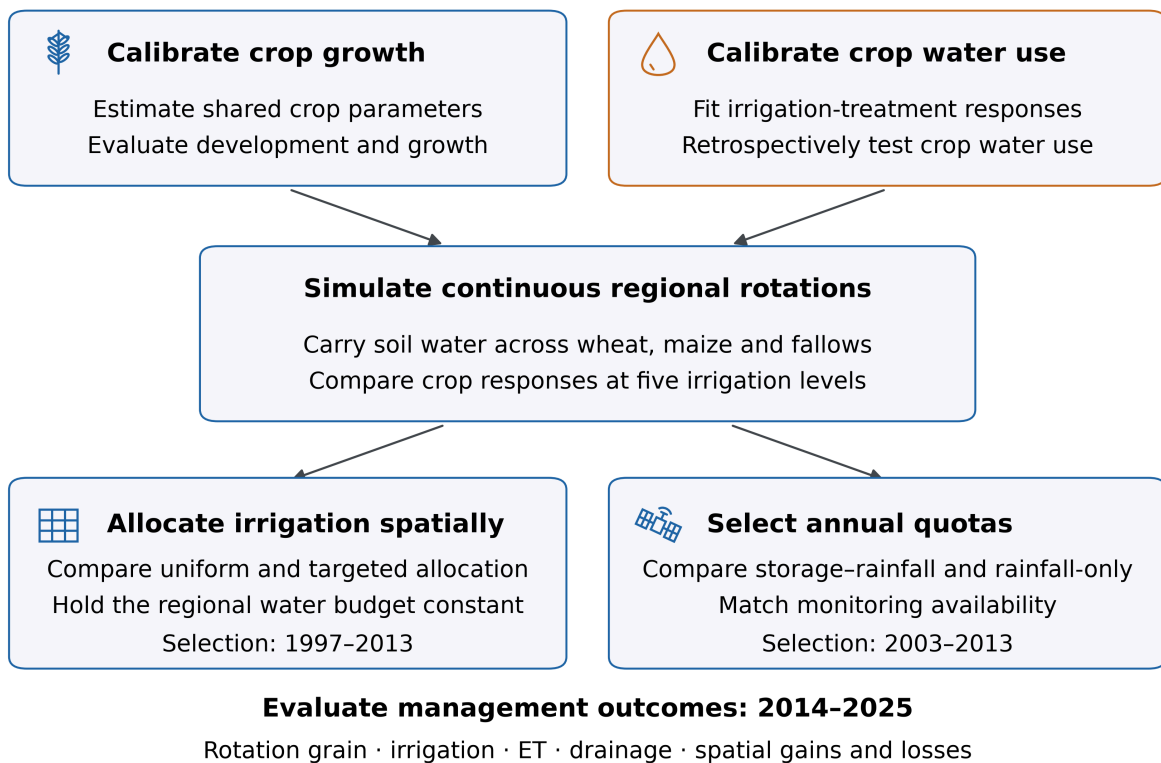


Figure 1. Study workflow. Crop growth and water-use calibration and evaluation precede continuous regional rotation simulations. Historical crop responses support spatial irrigation allocation under equal water budgets and annual quota selection from antecedent water availability. The strategies are compared during 2014–2025 using rotation grain production, field irrigation, crop-plus-fallow ET, profile drainage and the spatial distribution of production gains and losses.

Third, historical simulated responses support two complementary irrigation decisions: where to allocate a regional water budget and how much irrigation to supply in a given year. Spatial targeting and uniform reductions share identical regional irrigation budgets. Annual quota selection compares antecedent storage–rainfall classes with rainfall-only controls under matched production targets, candidate quotas and monitoring availability. Finally, the

selected strategies are compared during 2014–2025 using rotation grain production, field irrigation, crop-plus-fallow evapotranspiration (ET), profile drainage and the spatial distribution and persistence of production gains and losses.

2.2. Study area

The study area encompasses the winter-wheat–summer-maize lowlands of the North China Plain within 112–121.5°E and 32–41°N, extending from Beijing, Tianjin and Hebei through Shandong and Henan to the northern Anhui and Jiangsu plains (Figure 2) (Wu et al., 2019). The landscape consists of extensive alluvial lowlands, with piedmont plains along the Taihang and Yanshan mountains in the west and north and lower coastal plains toward the east (Long et al., 2025; Xu et al., 2026). Its low relief supports intensive agricultural production, while climatic and soil-water differences contribute to spatial variation in irrigation requirements (Ren et al., 2022).

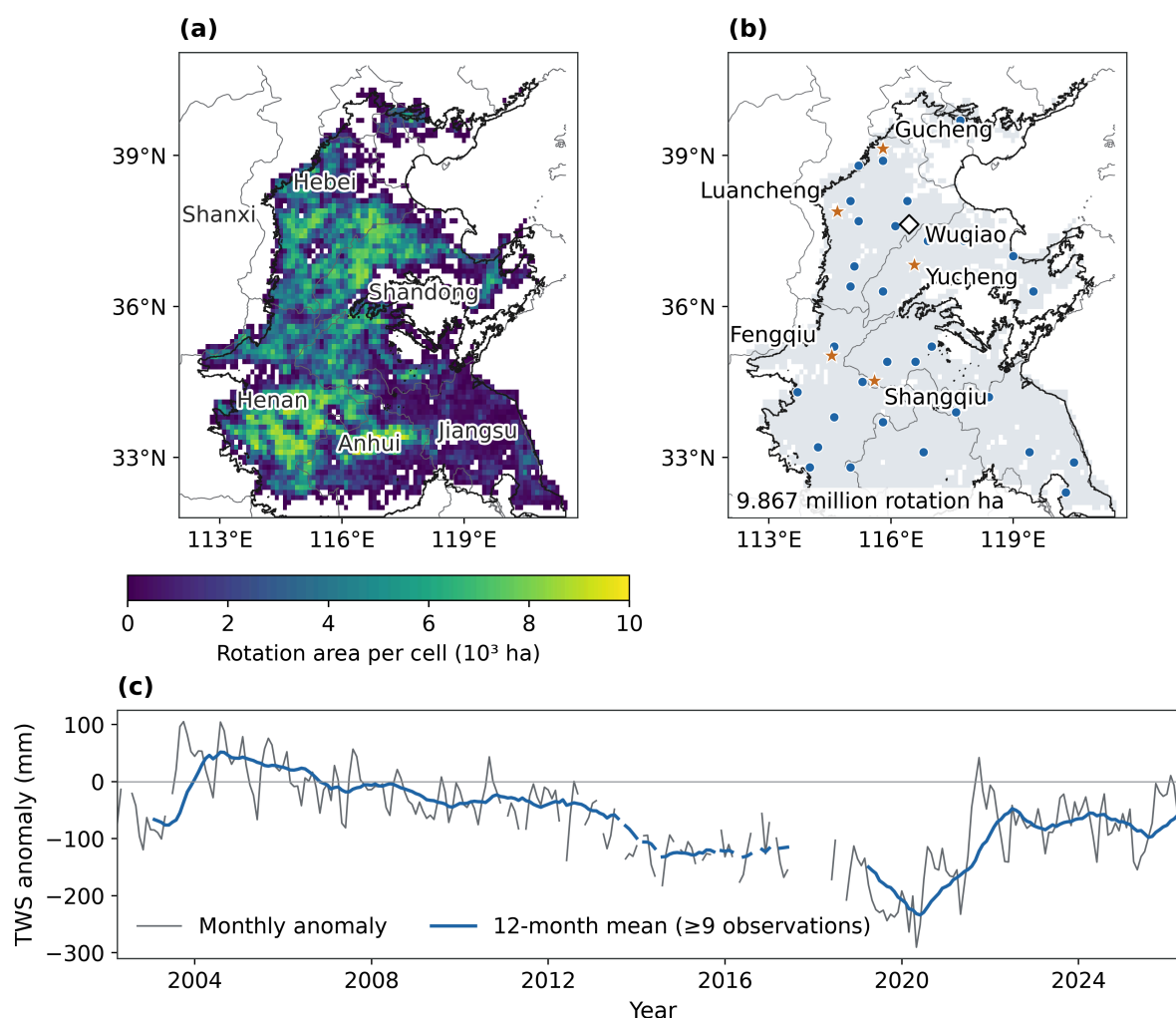


Figure 2. Study area, rotation area, observation sites and regional water storage. (a) Mapped 2020 wheat–maize rotation area. (b) Five crop-observation sites, the Wuqiao irrigation experiment and 32 regional simulation units. (c) Observed CSR monthly terrestrial-water-storage anomalies and a 12-calendar-month mean requiring at least nine observations. Missing months remain

gaps. Thin grey lines show provincial boundaries; black outlines delimit the operational physiographic study domain. Maps use an equirectangular projection (36°N, 117°E). Provincial boundaries: Natural Earth Admin 1 v5.1.1 (Natural Earth, n.d.).

The region has a continental monsoon climate, with cold, dry winters and warm, rainy summers; regional assessments report mean annual temperatures of approximately 10–15°C (Ren et al., 2022). Rainfall is concentrated in the summer monsoon, whereas spring is frequently dry and precipitation varies substantially among years and locations (Wu et al., 2019; Xu et al., 2026). Winter wheat is sown in autumn and harvested in early summer, followed by summer maize harvested in autumn. Wheat therefore grows largely outside the rainy season and commonly requires supplementary irrigation before sowing and during spring growth. Maize overlaps the rainy season, with irrigation demand depending more strongly on rainfall distribution and dry-year conditions. The two crops share a soil-water reserve, so irrigation in wheat can affect the water available to subsequent maize (Yang et al., 2024; Yang et al., 2022).

Groundwater has historically supplied much of the irrigation deficit, particularly in the northern plain, where intensive pumping for agriculture and urban demand caused prolonged groundwater decline (Xu et al., 2026). The aquifer system includes both unconfined and deeper confined units, with substantial spatial differences in groundwater depth and depletion (Long et al., 2025). Irrigation water availability also reflects access to river diversions and the allocation of transferred surface water. Since 2020, groundwater recovery has been documented across much of the northern North China Plain following the expansion of South-to-North Water Diversion supplies, reduced abstraction and water-saving measures, with wet years reinforcing recovery (Long et al., 2025; Xu et al., 2026). This transition places irrigation management within a changing water-supply setting: maintaining recovery requires attention to consumptive use and seasonal demand as well as the volume delivered to fields.

2.3. Field observations and environmental data

2.3.1. Multisite crop observations

Winter-wheat and summer-maize observations supported crop-growth parameter estimation and comparisons across five North China Plain sites. Long-term monitoring at Fengqiu, Luancheng, Shangqiu and Yucheng (Fengqiu Station, 2020a; Luancheng Station, 2020; Shangqiu Station, 2021; Yucheng Station, 2025a; Yucheng Station, 2021) was complemented by the Gucheng maize sowing-date experiment (Geng et al., 2023). The National Ecosystem Science Data Center archives contain intermittent crop seasons during 1998–2022; exact site–crop–year coverage and dataset identifiers are given in Tables S1 and S2.

Whole site-years retained their calibration or retrospective-evaluation assignments (Figure 3). The archive comprised 132 crop–plot–season trajectories, of which 127 contained eligible LAI, biomass, ET or harvest observations (Table 1). Daily ET and antecedent soil moisture came from the Fengqiu soil-water archive (Fengqiu Station, 2020b); ET was also available from the Yucheng water-environment archive (Yucheng Station, 2025b). Grain and harvest-biomass samples were confined to Yucheng. Within-season measurements and quadrats describe temporal dynamics and sampling variation rather than independent growing seasons.

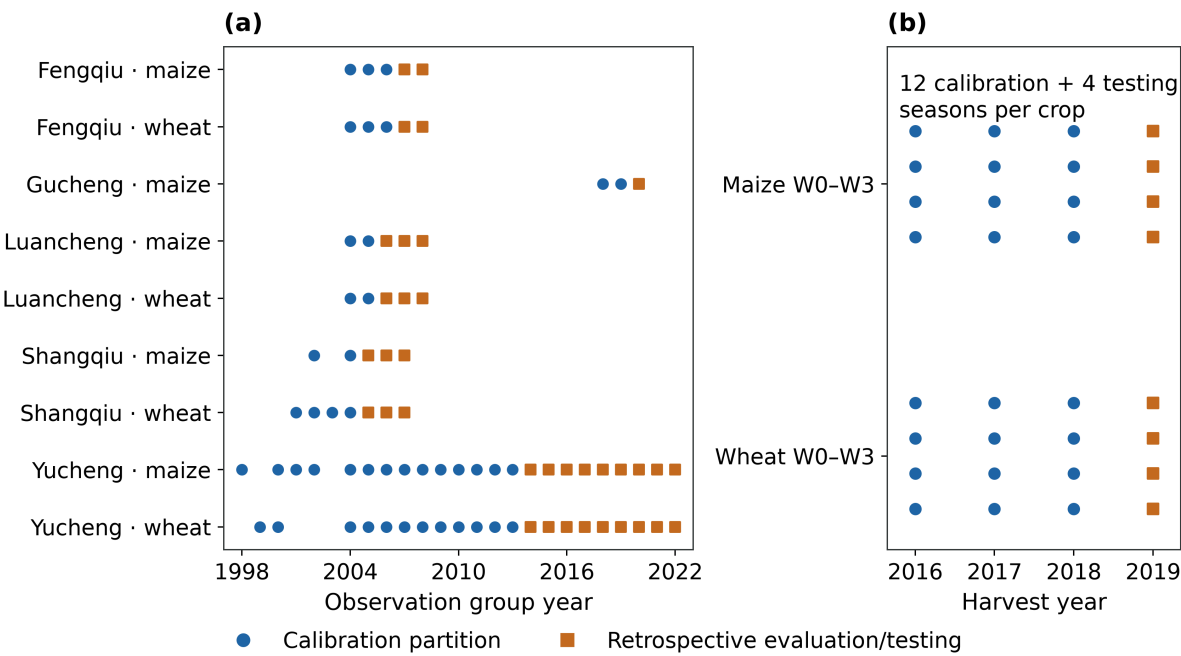


Figure 3. Evidence partitions. (a) Whole station site-years retain their original calibration and retrospective-evaluation partitions; dates denote observation-group years. All five eligible sites are included. (b) Four Wuqiao treatment means per crop-year provide 12 calibration seasons in 2016–2018 and four retrospective testing seasons in 2019. Station comparisons retain earlier growth-fitting history and unconfirmed complete irrigation management.

Observations were matched to predictions by crop, plot, quantity and sampling date or averaging window. Screening excluded wheat LAI above $7\text{ m}^2\text{ m}^{-2}$ and duplicated maize harvest groups of unresolved authenticity; raw values and exclusion reasons were retained. Complete plot-specific irrigation histories remain unconfirmed, so station comparisons are conditional on archived management assumptions and have no weight in the water-response fit based on documented Wuqiao treatments (Section 2.3.2). Measurement definitions, screening details and antecedent soil-moisture initialization are specified in Supplementary Section S1.

150 **Table 1. Eligible station-observation records by site and variable.**

Site	LAI	Biomass	Daily ET	Yield	Harvest biomass
Fengqiu	159	163	962	0	0
Gucheng	216	0	0	0	0
Luancheng	123	165	0	0	0
Shangqiu	114	148	0	0	0
Yucheng	183	67	4713	192	192

151 Counts denote individual observations, not independent experiments. Station comparisons retain inherited growth-
152 fitting history and unconfirmed complete management. Yucheng grain and harvest biomass contain six dry-matter
153 samples per crop-year; the duplicated 2012 and 2017 maize groups remain excluded.

154 **2.3.2. Field irrigation experiment**

155 The Wuqiao winter-wheat–summer-maize experiment (37.630°N, 116.444°E) comprised four wheat irrigation
156 regimes during 2015–2019 (Yang et al., 2024). W0 received no wheat irrigation; W1 received 75 mm before
157 sowing; W2 received an additional 75 mm at jointing; and W3 received a further 75 mm at anthesis. Total wheat
158 irrigation was therefore 0, 75, 150 and 225 mm, respectively. Treatments had three field replicates and a common
159 nitrogen input of 158.4 kg ha⁻¹ per crop.

160 Maize irrigation was identical across treatments within each year: no irrigation in 2016 and 2018, and 75 mm on 18
161 June in 2017 and 2019. Maize W0–W3 labels consequently denote preceding wheat irrigation rather than different
162 maize irrigation schedules. The treatment structure distinguishes direct wheat irrigation contrasts from maize
163 responses associated with preceding wheat management.

164 Crop–treatment–year observations comprised seasonal ET, grain yield and dry aboveground biomass. ET was
165 estimated from rainfall, irrigation and depletion of 0–2 m soil-water storage, assuming negligible runoff and
166 drainage; grain yields were standardized to 13% moisture (Yang et al., 2024). Observations from 2016–2018
167 constrained water-response coefficients, while 2019 remained outside fitting and parameter selection. Prior
168 inspection during model development makes the latter comparison retrospective (Figure 3). Simulation
169 initialization, ET accounting and grain digitization are detailed in Supplementary Section S4.

170 **2.3.3. Meteorological, soil, crop-rotation and terrain data**

171 Daily AgERA5 v2.0 weather data on the common 0.1° grid supplied minimum and maximum air
172 temperature, precipitation, solar radiation, reference evapotranspiration, vapor pressure and 10-m wind
173 speed for 1996–2025 (Copernicus Climate Change Service, 2020). Reference evapotranspiration (ET_0)
174 entered the water-balance module directly.

175 Native 250 m SoilGrids v2.0 maps of texture, bulk density, organic carbon and coarse fragments were averaged to
176 0.01° , then aggregated over mapped rotation pixels within 0.1° meteorological cells, for six depth intervals between
177 0 and 2 m (ISRIC — World Soil Information, 2020). Hydraulic properties were derived using Saxton–Rawls
178 equations with bulk-density and gravel adjustments (Saxton and Rawls, 2006). Organic matter was approximated
179 using an assumed organic-matter/organic-carbon ratio of 1.724, and air-dry water content was assumed to equal half
180 the wilting-point value. Hydraulic derivation is detailed in Supplementary Section S2.

181 The 500 m ChinaCP map for 2020 identified winter-wheat–summer-maize rotation pixels, which were paired with
182 the corresponding CPM cropland-percentage map (Qiu et al., 2022). Physical classified-cell areas were corrected
183 for projection scale and multiplied by cropland percentages to obtain rotation-area weights. CPM values were
184 sampled at ChinaCP pixel centers after coordinate transformation because the source grids had different origins.
185 Pixel centers assigned these weights to the study area and 0.1° meteorological cells. The rotation footprint was
186 9,866,733 ha before soil screening; three cells with wholly missing soil profiles, totaling 131 ha, were excluded,
187 leaving 9,866,602 ha in 3,641 meteorological cells. The fixed 2020 footprint supplies a common area basis across
188 climate years, so modeled totals describe climatic and management responses rather than historical production
189 trends. The 2015–2021 area series and common-extent comparison with CCD2020 quantify mapping sensitivity in
190 Supplementary Section S3.

191 SRTMGL1 v003 elevation was bilinearly resampled to 0.001° , and slope was derived at that support (NASA JPL,
192 2013). Aligned 10×10 cell means provided elevation and slope on a 0.01° grid. Catchment screening, elevation
193 ≤ 200 m and slope $\leq 2^\circ$ delineated a connected lowland study area of 404,658 km² within the geographical extent
194 described in Section 2.2. Geographical and terrain-threshold sensitivities are reported in Supplementary Section S3.
195 Crop-area weights and aggregated soil profiles were linked to daily weather through common 0.1° cell identifiers.

196 **2.3.4. Terrestrial-water-storage observations**

197 CSR RL06.3 monthly GRACE/GRACE-FO total-water-storage anomalies were aggregated using physical cell
198 areas and the official CSR land mask (Save, 2020; Save et al., 2016). The land-mask overlap covers 395,969 km²
199 within the 404,658-km² study area; excluded coastal supports account for 2.15% of its area. The archive contains
200 259 observed monthly solutions from April 2002 to July 2026 relative to the 2004–2009 mean. Source values in
201 centimeters were converted to millimeters, and missing months remain missing. A 12-calendar-month moving mean
202 requires at least nine observations. The 0.25° product grid is finer than the effective GRACE observation scale, so
203 one regional storage indicator is shared across all simulation units (Save, 2020; Save et al., 2016). Soil, surface and
204 other stores were not subtracted; the indicator represents total water storage rather than groundwater alone.
205 The corresponding aggregation without the CSR land mask is retained as a sensitivity analysis. Regional storage
206 changes provide hydrological context for water-management pressures, with no causal attribution to particular
207 irrigation treatments, water transfers or groundwater regulations (Long et al., 2025; Long et al., 2020).

208 **2.4. Crop model description**

209 Open Crop Model (OCM) (Zhao et al., submitted) couples crop growth and a layered soil-water balance at a daily
210 time step (Figure 4). Wheat development combines temperature, vernalization and photoperiod responses; maize
211 development follows thermal time. Development stage (DVS) is anchored at emergence, flowering and maturity.
212 Germination begins when water above the wilting-point storage is available in the represented layer containing the
213 seed; subsequent thermal development starts from that date. Stage coefficients remain fixed, while the soil-water
214 state can alter establishment timing. Recorded or prescribed harvest dates define the management endpoint
215 separately. Soil-water storage passes continuously through crops and fallows.

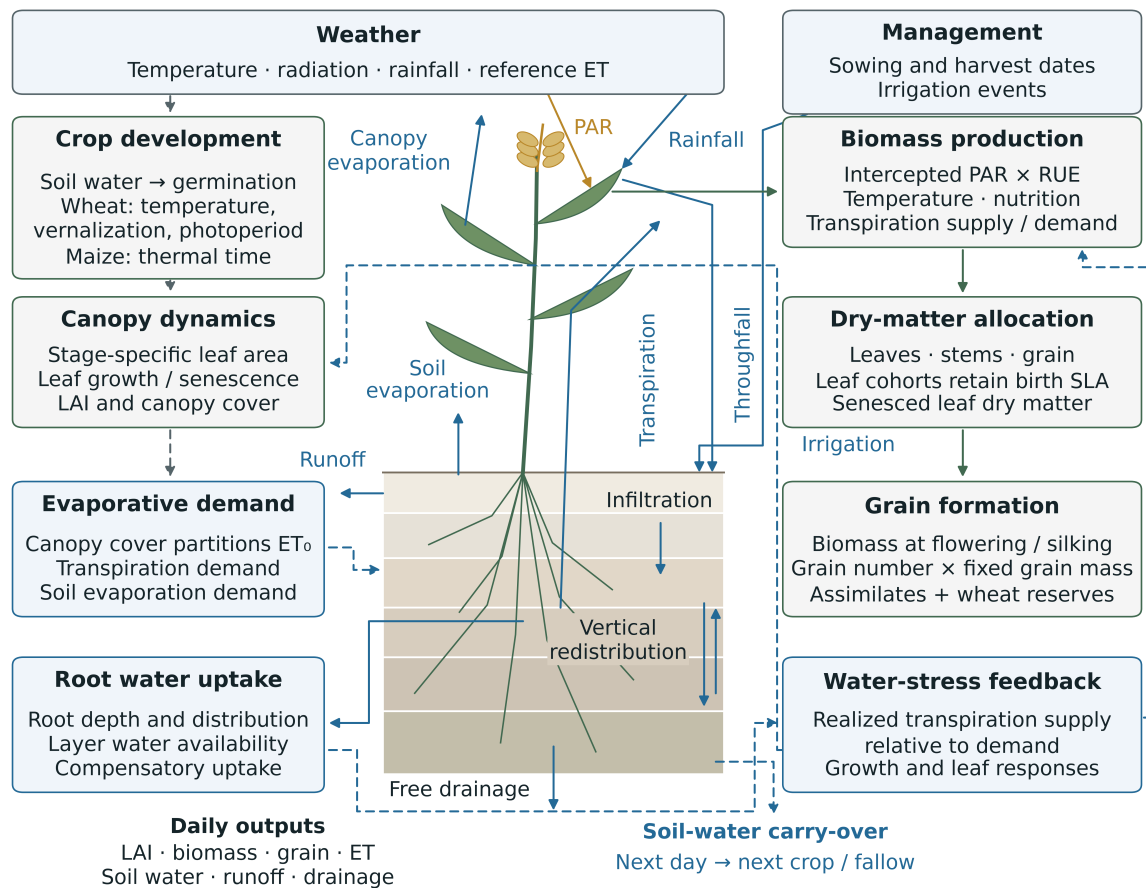


Figure 4. Crop growth and soil-water coupling in Open Crop Model. Gold denotes photosynthetically active radiation (PAR) input; blue solid arrows denote water fluxes; green solid arrows denote dry-matter transfers; dashed arrows denote process controls and feedbacks. Nutrition is prescribed, the lower boundary uses free drainage, and soil water carries through crop/fallow segments. Plant and soil sketches are not to scale.

Canopy interception of radiation follows Beer–Lambert attenuation. Aboveground production depends on intercepted PAR, crop-specific RUE, temperature, prescribed nutrition and an empirical power response to realized transpiration supply. PAR is 0.45 of incident shortwave radiation (Kiniry et al., 1989). Stage-specific SLA determines new leaf area, while established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Stage-dependent thermal senescence and a separate daily stress-loss fraction determine leaf survival (Tan et al., 2020). LAI emerges from these carbon and area balances without a maximum-LAI parameter. Simulated biomass at wheat flowering or maize silking determines grain number; fixed individual grain mass defines sink capacity, and assimilate supply and wheat stem reserves constrain grain filling.

The water balance represents interception, ponding, infiltration, evaporation, root uptake, runoff and bottom drainage. Atmospheric demands are partitioned from reference ET using canopy cover and effective transpiration and evaporation coefficients. Soil evaporation follows a storage-based two-stage reduction (Allen et al., 1998).

231 Root uptake depends on rooted-layer water, local depletion stress and a bounded compensation fraction that
232 redirects unmet demand toward better-supplied rooted layers. Vertical transport uses van Genuchten–Mualem
233 hydraulic functions (Mualem, 1976; van Genuchten, 1980), signed gravity–matric gradients and conservative
234 storage bounds. Relative-arithmetic interface mobility and adaptive internal steps permit wetting and redistribution
235 within 24 daily outer steps. The lower boundary is free drainage with no capillary-rise supply. This explicit layered
236 approximation differs from a full implicit Richards-equation solution. Profile depths follow the supplied inputs,
237 including 2 m for Wuqiao and regional simulations.

238 **2.5. Model parameterization, calibration and evaluation**

239 Phenological coefficients, stage-specific specific leaf area and senescence, initial LAI, stem-reserve coefficients and
240 nutrition modifiers follow the crop parameterization described in Section S2. RUE, leaf allocation and grain-
241 number parameters were estimated from 480 wheat and 433 maize growth and harvest observations across four
242 wheat sites and five maize sites. Whole site-years retain their original partitions, with balanced contributions from
243 variables, sites, site-years and crop cases. Station simulations retain their recorded site and cultivar
244 parameterization; Wuqiao and regional simulations use the shared crop parameters (Table S3). Individual grain
245 masses are fixed at 0.045 g for wheat and 0.300 g for maize.

246 Eight water-use parameters per crop are fitted to the documented Wuqiao treatments in 2016–2018: the
247 transpiration coefficient, soil evaporation coefficient, readily available water fraction, assimilation water-stress
248 exponent, root-density decay, maximum rooting depth, root water-uptake coefficient and root compensation
249 fraction. Each crop contributes 12 seasonal ET, 12 dry-grain and 12 dry-biomass targets, with three annual W3–W0
250 ET contrasts derived from the same treatment seasons. RUE, leaf allocation and grain-number parameters remain
251 fixed. Bounded least squares combines scaled observation residuals and weak parameter priors; objective weights,
252 bounds and convergence settings are specified in Section S2.4.

253 The fitted water parameters are fixed before comparison with the four 2019 treatment means per crop. Previous
254 inspection of these outcomes makes the test retrospective. Station observations have zero weight in this water fit
255 and assess transfer under unconfirmed complete irrigation histories. Regional strategy selection uses historical
256 weather responses; the crop-calibration years overlap the 2014–2025 regional comparison period, so those
257 outcomes are conditional scenarios rather than time-independent crop validation. Observed and simulated quantities
258 retain matched sampling windows and grain-moisture bases.

259 **2.6. Regional upscaling and irrigation scenarios**

260 Thirty-two area-weighted climate–soil representatives were selected from actual source cells using deterministic
261 clustering of latitude, longitude, plant-available water, and 1997–2013 mean precipitation and reference ET.
262 Features were standardized with rotation-area weights. Each cluster uses the actual cell nearest its centroid; the full
263 source-cell mapping and represented areas are retained. Climate years used for policy evaluation do not enter the
264 climate features used for selection. Shared crop parameter sets are transferred regionally without extrapolating site-
265 specific adjustments.

266 Each representative has five continuous histories at irrigation fractions 0, 0.25, 0.50, 0.75 and 1.00. The
267 conventional wheat schedule is 60 mm at sowing on 12 October, 90 mm on 25 March, 90 mm on 15 April and 60
268 mm on 10 May, totaling 300 mm. Maize receives 80 mm at sowing on 15 June. These common schedules are
269 declared scenarios and are not observed universal practice across the plain. Every event amount is multiplied by the
270 irrigation fraction, with its timing fixed. Wheat is simulated through 10 June and maize through 5 October;
271 intervening fallows carry the soil state forward. Initial plant-available water is 80% at the start of the 1996 spin-up,
272 with no subsequent seasonal storage reset. All crop and fallow ET, drainage and runoff are included. The final 2025
273 cycle ends on 5 October because the next fallow lies outside the declared endpoint. Nutrition is represented by
274 prescribed fertilized-crop modifiers; nitrogen cycling and spatially varying nutrient limitations are not dynamically
275 simulated.

276 Eight purposively selected non-representative cells provide paired spatial aggregation checks under all five
277 irrigation fractions. Two cells from each latitude quartile have the largest area-times-feature-distance score. Actual-
278 cell weather and profiles use the same continuous model and crop parameters. Their differences from
279 representative-cell predictions measure crop-model aggregation error rather than field-observation accuracy
280 (Supplementary Section S3).

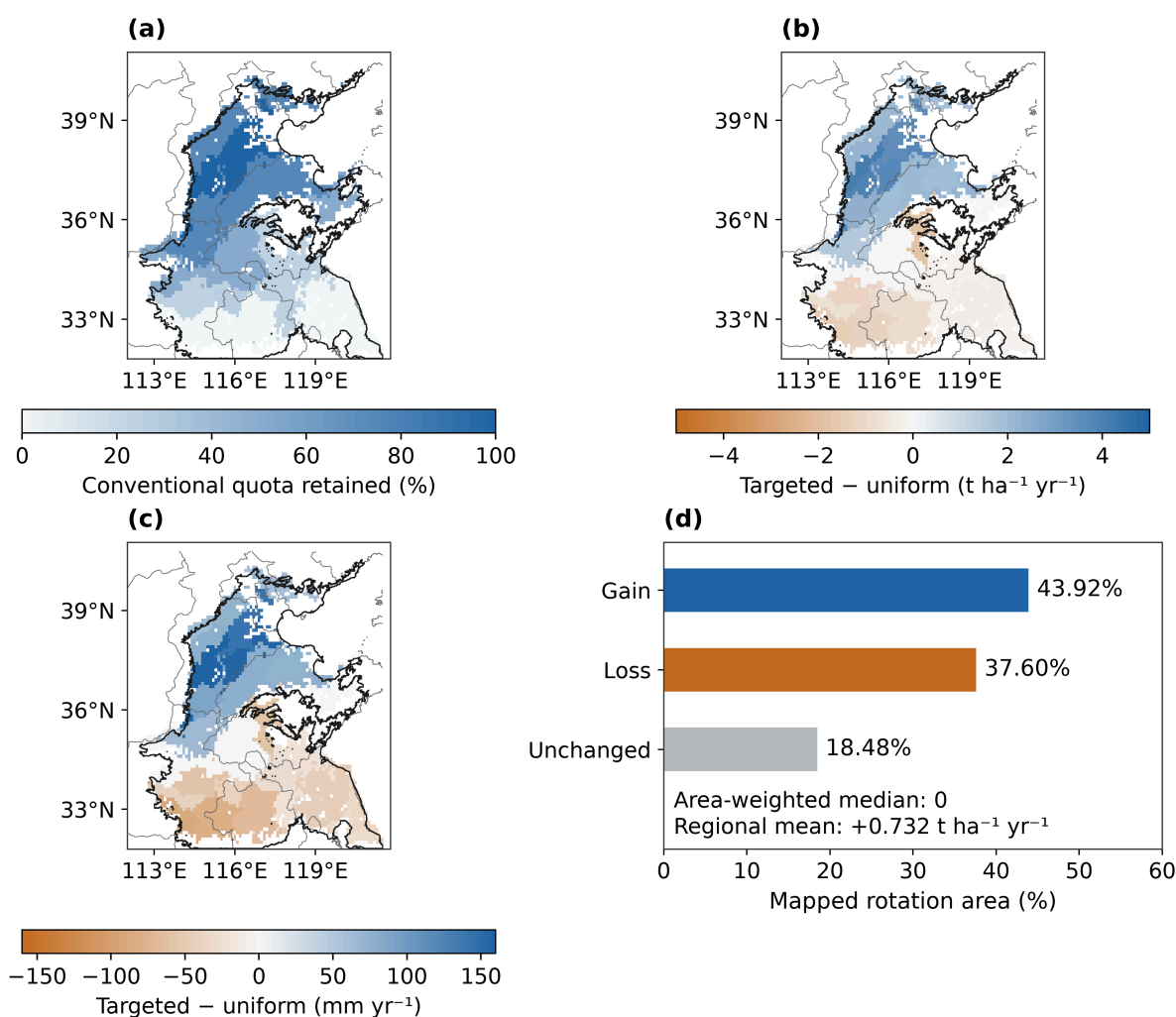
281 **2.7. Irrigation strategies and comparisons**

282 **2.7.1. Spatial allocation under common irrigation budgets**

283 Uniform allocation assigns the same irrigation fraction to every represented hectare. Targeted allocation
284 chooses area shares x_{jk} for representative j and simulated irrigation fraction k by linear programming. The
285 objective maximizes $\sum_j A_j \sum_k x_{jk} \bar{Y}_{jk}$, where $\bar{Y}_{jk}$ is mean annual wheat-plus-maize grain in 1997–2013.
286 Constraints require $\sum_k x_{jk} = 1$, $0 \leq x_{jk} \leq 1$ and $\sum_j A_j \sum_k x_{jk} I_k = (1 - r) \sum_j A_j I_1$, with reductions

287 $r = 0.25, 0.50$ or 0.75 . Within-representative mixtures describe shares of area receiving actual simulated
 288 irrigation treatments; crop responses are not interpolated between irrigation fractions. Allocations remain
 289 fixed in 2014–2025. Both policies deliver exactly the same total field-irrigation water in every year, so
 290 their differences reflect allocation under the same volume constraint.

291 Spatial contrasts at the common 50% irrigation cut use frozen shares of complete continuous irrigation histories.
 292 Their area-weighted grain, ET and drainage differences map from 32 representatives to 3,641 source cells. Fixed
 293 southern ($32\text{--}36^\circ\text{N}$), central ($36\text{--}38^\circ\text{N}$) and northern ($38\text{--}41^\circ\text{N}$) bands summarize the distribution. Counts of
 294 positive annual contrasts describe consistency within the modeled record (Figure 5; Table S9).



295 **Figure 5. Spatial redistribution at a common 50% regional irrigation reduction, 2014–2025. (a) Targeted fraction of the conventional**
 296 **quota; uniform allocation retains 50% everywhere. (b,c) Targeted-minus-uniform combined rotation dry grain and crop-plus-fallow**
 297 **ET. (d) Mapped rotation-area shares with positive, negative or unchanged mean grain contrasts. The 3,641 cells inherit 32**
 298 **representative responses. Thin grey provincial lines and black study-domain outlines follow Figure 2; unmapped areas remain blank.**

299 **2.7.2. Annual quotas based on antecedent water availability**

300 Relative water-availability classes combine two antecedent indicators: regional total-water-storage status and local
301 rainfall before wheat sowing. The storage indicator is the mean of available August and September GRACE
302 observations in the calendar year preceding harvest, with at least one monthly solution required. Precipitation is
303 accumulated from 1 July to 30 September at each simulation unit, covering 92 days before the prescribed 12
304 October wheat sowing. The corresponding precipitation/reference-ET ratio is retained as a diagnostic of climatic
305 water availability. These antecedent indicators do not use rainfall or crop outcomes from the subsequent growing
306 season.

307 Thresholds use 2003–2013 harvest years, the common selection period with antecedent GRACE coverage. The
308 regional storage median is 13.38 mm, and the rotation-area-weighted precipitation median is 431.74 mm. Values
309 below their respective thresholds define lower storage and lower rainfall; values at or above define higher storage
310 and higher rainfall. Crossing the two indicators gives four hydroclimatic classes. The classes describe relative
311 antecedent water availability, rather than absolute water scarcity derived from water demand, allocation rights or
312 access. Spatial differences within a year arise from precipitation; GRACE does not supply independently resolved
313 field-scale storage classes.

314 For each class, five continuous irrigation histories provide paired grain and ET responses relative to conventional
315 irrigation, weighted by rotation area and year. The primary candidate is the lowest fraction retaining at least 95% of
316 class-mean conventional grain in 2003–2013; a 98% target assesses a stricter production criterion. Selected
317 fractions and class thresholds remain frozen in 2014–2025. Higher-storage classes have no evaluation exposure.
318 Required GRACE observations are missing for 2014, 2018 and 2019.

319 Two annual strategies apply the irrigation fractions selected from fixed-fraction responses at grain-retention targets
320 of 95% and 98%. At each simulation unit, the antecedent storage–rainfall class determines the fraction before wheat
321 sowing. This fraction scales all four wheat applications and the subsequent maize application in the same rotation
322 year. Application dates, crop parameters, weather, hydraulic profiles and the free-drainage lower boundary are
323 identical to those in the fixed-fraction experiment. Conventional irrigation applies before 2003 and whenever an
324 eligible storage class is unavailable, including evaluation harvest years 2014, 2018 and 2019.

325 Each strategy is simulated continuously from the 1996 spin-up through 2025. Soil-water states pass between crops
326 and fallows and across annual changes in irrigation fraction. Conventional irrigation supplies the paired continuous
327 reference; replays at two simulation units verify the scenario driver against conventional outputs. Class thresholds
328 and irrigation fractions remain fixed after selection. Outcomes cover all twelve comparison years and, separately,

the nine years with eligible classes. Production retention is the ratio of period-mean regional grain production to its corresponding conventional mean; annual ratios describe variation within the period. The strategies assume access to antecedent observation periods, while release availability of GRACE before sowing remains unverified.

2.7.3. Rainfall-only controls and matched monitoring coverage

Rainfall-only controls use the same 2003–2013 simulation-unit years with eligible GRACE observations, the same rotation-area weights, irrigation candidates and grain-retention targets. Selection groups responses by antecedent rainfall alone and chooses the lowest fraction meeting the relevant target. The rainfall threshold is identical to that in the joint classification. Matched controls receive conventional irrigation whenever the storage–rainfall strategy lacks an eligible class, preserving both the selection cohort and monitoring coverage. Continuous differences between these paired strategies consequently assess the incremental effect of storage-based grouping.

Separate rainfall-available controls apply the selected rainfall-only fractions whenever an eligible rainfall group exists, including years without GRACE coverage. Their complete histories assess the combined effect of rainfall-based management and greater monitoring availability. Irrigation changes in earlier gap years can alter the soil-water state of later crops, so these controls are reported separately from the matched comparison.

2.8. Evaluation metrics, spatial statistics and water accounting

Performance is summarized by RMSE, mean bias error, nRMSE relative to observed mean, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and squared Pearson correlation (R^2). Station scores give equal weight to available sites, site-years within sites, crop cases within site-years and records within cases. Wuqiao treatment means receive equal weights within each crop and partition. Field calibration and retrospective testing remain separate. Grain and harvest biomass in station comparisons are dry matter; Wuqiao grain results retain the published 13% moisture basis, with a single 0.87 conversion between dry and reported grain. Biomass remains dry matter throughout.

Seasonal curves group matching observations and predictions into 15-day wheat and 7-day maize bins at observation-window midpoints in days after sowing. Case means receive equal weight within site-years, site-year means within sites and available sites within bins. Standard errors describe variation among contributing site-year means rather than model uncertainty. Site-specific curves use equal site-year weights. Coverage varies among bins, and gaps remain uninterpolated. Accuracy scores use unbinned records. Yucheng harvest error bars describe quadrat sampling variation within crop-years.

ET sums use identical observed and simulated measurement days within each case and retain the station balancing weights. They exclude unmeasured days and do not represent complete-season observed totals. Regional grain production uses dry mass and rotation-area weights; actual ET includes crops and fallows. Paired policy differences cover 2014–2025. Daily water and carbon residuals and continuity of carried soil-water storage assess numerical conservation. Field irrigation, ET, drainage below 2 m and groundwater storage have distinct accounting boundaries.

Spatial distribution statistics describe targeted-minus-uniform policy contrasts in mean annual combined wheat–maize dry grain and water fluxes over 2014–2025. The 3,641 mapped 0.1° source cells carry outcomes from 32 representative response units. Cell weights are the static 2020 mapped rotation areas after cropland-fraction correction; they represent mapped rotation area rather than observed harvested area. Gains and losses exceed $+10^{-6}$ and fall below -10^{-6} kg ha $^{-1}$, respectively; contrasts within these bounds are classified as unchanged. Group means use rotation-area weights, regional contributions sum area multiplied by the corresponding contrast, and weighted quantiles use the inverse empirical cumulative area distribution. Annual loss frequencies use the same sign threshold across the 12 years. Connected loss footprints comprise occupied cells sharing a horizontal or vertical grid edge (rook adjacency). The statistics describe the conditional modeled allocation footprint; mapped cells do not provide independent replication or cell-specific validation.

3. Results

3.1. Conditional multisite crop comparisons

Retrospective-evaluation LAI RMSE was 1.843 m 2 m $^{-2}$ for wheat and 0.995 m 2 m $^{-2}$ for maize, with NSE of 0.184 and 0.689, respectively (Figure 6). LAI RMSE in the original calibration partition was 1.727 and 1.138 m 2 m $^{-2}$. In-season biomass RMSE in retrospective evaluation was 4.413 and 2.789 t ha $^{-1}$, with biases of +1.314 and +1.158 t ha $^{-1}$ and NSE of 0.349 and 0.796. Site-specific seasonal comparisons retain their available observations and original partitions (Figures S6–S11).

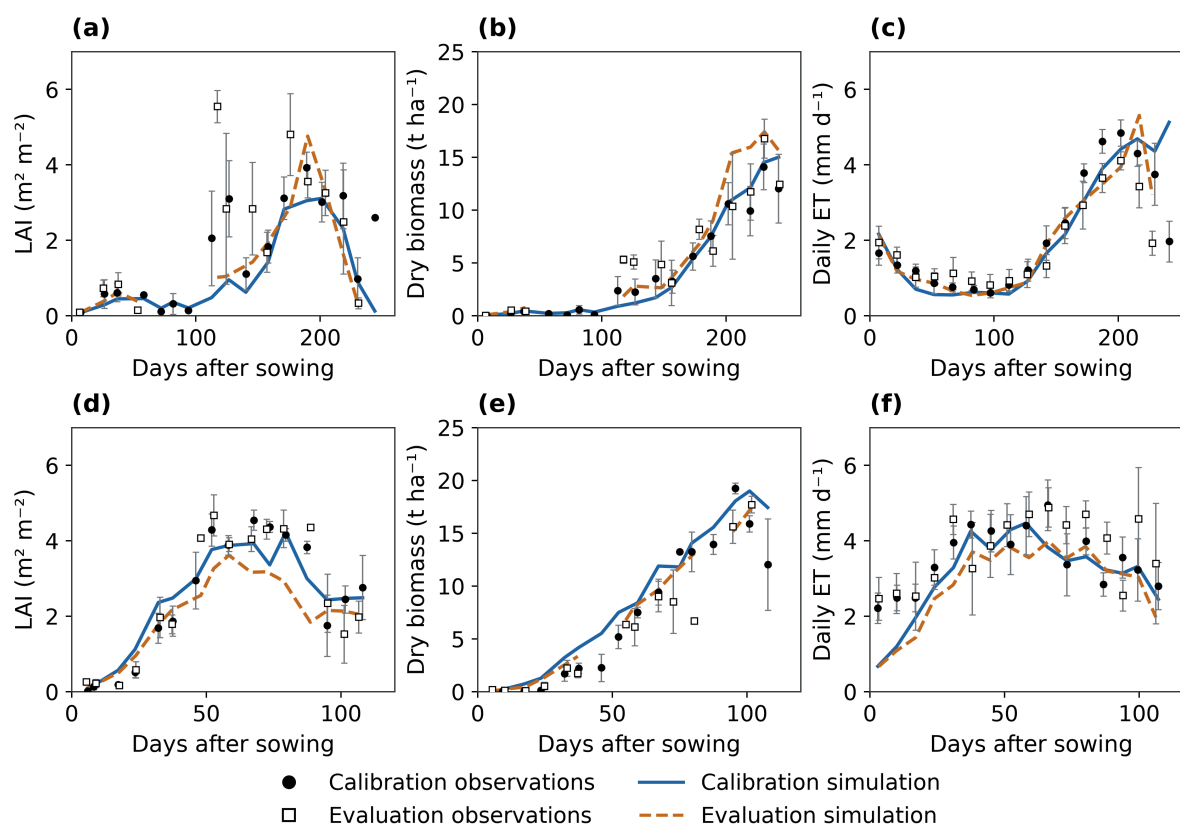


Figure 6. Multisite seasonal crop comparisons. Wheat and maize occupy the top and bottom rows; columns show LAI, aboveground dry biomass and daily actual ET. Filled and open observations distinguish the original calibration and retrospective-evaluation partitions; solid blue and dashed orange lines show the corresponding simulations. Error bars describe standard errors among site-year means. Balanced means use 15-day wheat and seven-day maize bins, and unobserved bins remain gaps. Accuracy scores use unbinned records. Complete station irrigation management remains unconfirmed.

Dry-grain RMSE was 1.256 t ha^{-1} for wheat and 1.402 t ha^{-1} for maize in the original calibration partition, compared with 0.834 and 2.286 t ha^{-1} in retrospective evaluation (Figure S5). Evaluation grain biases were $+0.485$ and -1.586 t ha^{-1} , and NSE was 0.099 and -1.566 . Evaluation harvest-biomass RMSE was 2.736 t ha^{-1} for wheat and 3.719 t ha^{-1} for maize; the corresponding observation counts were 54 in nine wheat years and 48 in eight maize years.

Daily ET RMSE in retrospective evaluation was 1.809 mm d^{-1} for wheat and 2.572 mm d^{-1} for maize, with biases of $+0.141$ and -0.734 mm d^{-1} and NSE of 0.121 and 0.012 (Figure 6). ET summed over matched measurement days had evaluation RMSE of 51.29 mm for wheat and 74.17 mm for maize, biases of $+25.94$ and -55.58 mm , and NSE of 0.826 and 0.118 (Table 2). These sums exclude days without ET measurements.

394 **Table 2. Conditional station ET sums on matched measurement days. Observed and simulated sums use identical measurement days**
395 **within cases, with balanced site, site-year and case weights. Unmeasured days are excluded. Original calibration partitions retain an**
396 **earlier growth-fitting history; evaluation is retrospective. Complete irrigation-log matching remains unconfirmed.**

Crop	Dataset	Seasons	NSE	RMSE (mm)	MBE (mm)
Maize	Prior calibration partition	11	0.511	45.60	-34.92
Maize	Retrospective evaluation	10	0.118	74.17	-55.58
Wheat	Prior calibration partition	11	-0.300	84.94	-23.53
Wheat	Retrospective evaluation	11	0.826	51.29	25.94

397 **3.2. Wuqiao calibration and retrospective testing**

398 Seasonal ET calibration RMSE was 40.19 mm for wheat and 37.44 mm for maize across 12 treatment seasons per
399 crop, with biases of -12.65 and $+3.60$ mm and nRMSE of 8.97% and 12.69% (Table 3; Figure 7). Across the four
400 2019 testing means per crop, ET RMSE was 73.80 and 55.64 mm, bias was $+48.93$ and -35.90 mm, nRMSE was
401 21.98% and 14.13%, and NSE was 0.503 and -0.346 , respectively.

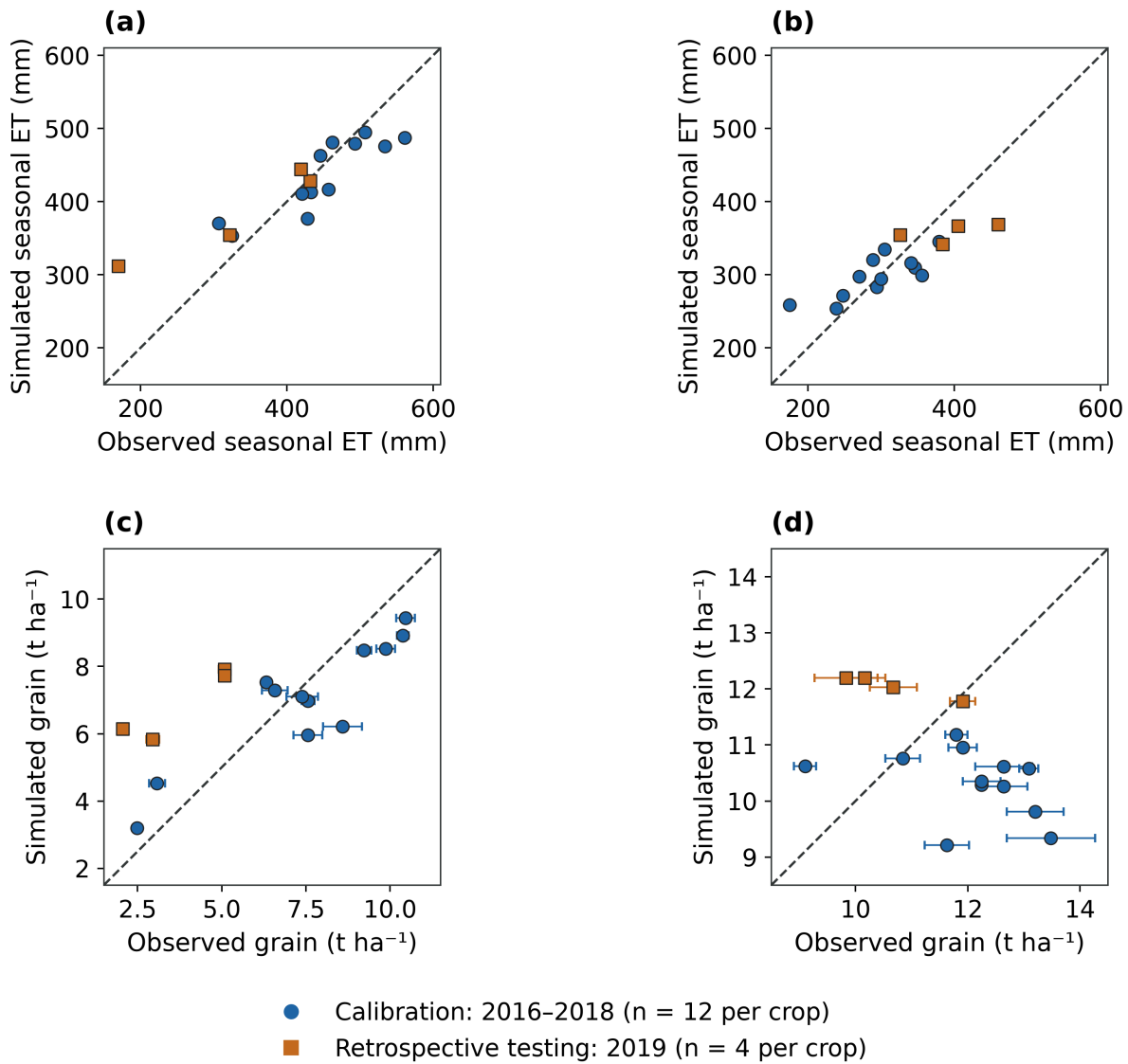


Figure 7. Wuqiao calibration and retrospective testing. (a,b) Seasonal ET; (c,d) annual treatment-mean grain at 13% moisture. Wheat occupies the left column and maize the right. Circles show 2016–2018 treatments used to fit water-use parameters; squares show 2019, withheld from the fit and selection. Dashed lines denote 1:1 agreement. Crop-specific grain axes focus on their observed ranges. Horizontal grain error bars are the published SE from three field replicates. Observed ET follows the published 0–2 m water balance; simulated dry grain is converted once by division by 0.87. Maize labels retain preceding wheat management with identical maize irrigation within each year.

At 13% grain moisture, calibration RMSE was 1.249 t ha⁻¹ for wheat and 2.267 t ha⁻¹ for maize; retrospective-testing RMSE was 3.161 and 1.700 t ha⁻¹. Testing grain biases were +3.108 and +1.403 t ha⁻¹, with nRMSE of 83.27% and 15.97% and NSE of -4.654 and -3.641 (Table 3; Figure 7). Dry aboveground-biomass RMSE was 4.222 and 4.688 t ha⁻¹ in calibration and 5.427 and 2.038 t ha⁻¹ in testing, respectively. Testing biomass nRMSE was 55.66% for wheat and 10.44% for maize.

W3–W0 seasonal ET-contrast RMSE was 45.79 mm for wheat and 66.84 mm for maize across three calibration years; the absolute 2019 contrast errors were 116.37 and 52.74 mm. For grain contrasts of W1–W3 relative to same-year W0, calibration RMSE was 0.519 and 1.669 t ha⁻¹ and retrospective-testing RMSE was 1.316 and 1.323 t ha⁻¹, respectively, at 13% moisture (Table S5; Figure S2). Maize treatment labels denote preceding wheat irrigation; maize irrigation was identical within each year.

Table 3. Wuqiao water calibration and retrospective testing.

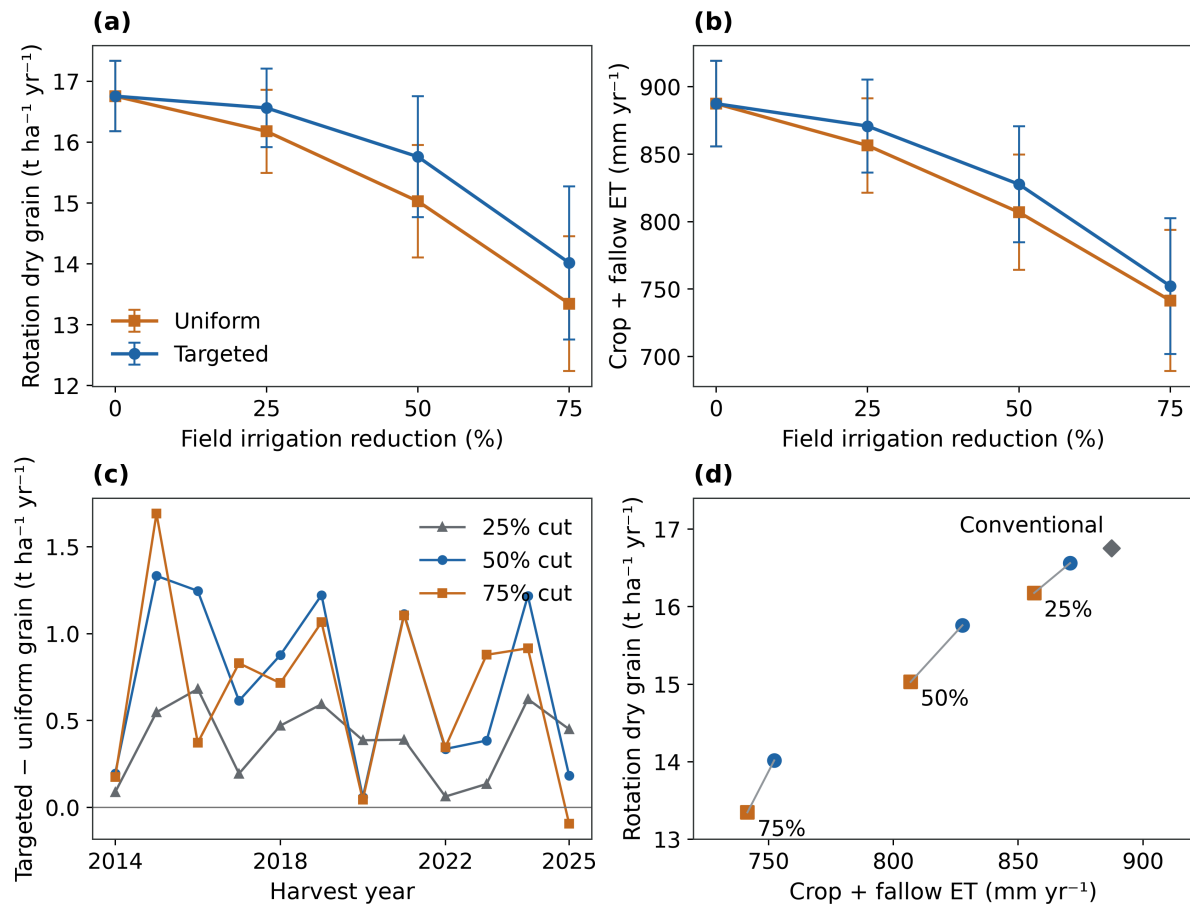
Crop	Period	Target	n	Unit	RMSE	MBE	nRMSE (%)	NSE	R ²
Maize	2016–2018	Seasonal ET	12	mm	37.444	3.604	12.69	0.533	0.619
Maize	2016–2018	Grain	12	t ha ⁻¹	2.267	-1.738	18.78	-2.940	0.128
Maize	2016–2018	Biomass	12	t ha ⁻¹	4.688	-4.188	21.79	-5.625	0.073
Maize	2019	Seasonal ET	4	mm	55.637	-35.902	14.13	-0.346	0.343
Maize	2019	Grain	4	t ha ⁻¹	1.700	1.403	15.97	-3.641	0.972
Maize	2019	Biomass	4	t ha ⁻¹	2.038	0.726	10.44	-0.313	0.123
Wheat	2016–2018	Seasonal ET	12	mm	40.188	-12.649	8.97	0.689	0.758
Wheat	2016–2018	Grain	12	t ha ⁻¹	1.249	-0.445	16.74	0.746	0.817
Wheat	2016–2018	Biomass	12	t ha ⁻¹	4.222	-2.480	25.48	0.496	0.781
Wheat	2019	Seasonal ET	4	mm	73.805	48.929	21.98	0.503	0.915
Wheat	2019	Grain	4	t ha ⁻¹	3.161	3.108	83.27	-4.654	0.870
Wheat	2019	Biomass	4	t ha ⁻¹	5.427	4.908	55.66	-0.752	0.924

Calibration uses 12 treatment seasons per crop in 2016–2018; retrospective testing uses four in 2019. Counts are treatment means rather than independent replicate seasons. ET is a 0–2 m water-balance estimate under negligible runoff and drainage assumptions. Grain is at 13% moisture and aboveground biomass is dry matter. Annual grain means and SE are digitized from source Supplementary Figure S2. Initial total storage is observed; layer distribution and transferred hydraulics remain conditional.

3.3. Grain production and water balance under irrigation allocation

At a 50% regional irrigation reduction, targeted allocation gave a simulated annual rotation dry-grain yield of 15.761 t ha⁻¹ yr⁻¹ for wheat and maize combined, compared with 15.029 under uniform cuts. Both yields remained below the conventional yield of 16.756 t ha⁻¹ yr⁻¹. Targeting therefore retained 0.732 t ha⁻¹ yr⁻¹ more grain than

428 uniform allocation (4.87% of uniform production), equivalent to 7.219 million t yr⁻¹ across the fixed 9.867 million
 429 rotation hectares (Table 4; Figure 8). Targeted production remained 9.813 million t yr⁻¹ below conventional
 430 production.



431 **Figure 8. Irrigation allocation and grain–water trade-offs during 2014–2025. (a,b) Rotation-area-weighted annual combined wheat–**
 432 **maize dry grain and crop-plus-fallow ET; error bars are between-year SD. (c) Annual targeted-minus-uniform grain at identical**
 433 **irrigation budgets. (d) Period means; connectors join policies with the same budget. Values per hectare use the same 9.867 million**
 434 **mapped rotation hectares, rather than separate crop areas. Targeting retains more grain than uniform cuts while remaining below**
 435 **conventional production.**

436 Both allocations supplied 190 mm yr⁻¹ of field irrigation. Targeted and uniform crop-plus-fallow ET averaged
 437 827.70 and 806.94 mm yr⁻¹, respectively. The grain advantage was accompanied by 20.76 mm yr⁻¹ (2.048 km³ yr⁻¹)
 438 greater ET. Profile drainage was 79.04 versus 96.34 mm yr⁻¹, and runoff was 14.43 versus 18.99 mm yr⁻¹ (Table 4).
 439 The targeted-minus-uniform grain contrast at the 50% cut was positive in all twelve comparison years (Figure 8c).
 440 The positive regional contrast was accompanied by spatially contrasting yield responses (Section 3.4).

441 **Table 4. Simulated annual rotation production and water balance under equal irrigation reductions, 2014–2025.**

Strategy	Dry grain (t ha ⁻¹ yr ⁻¹)	Irrigation (mm yr ⁻¹)	ET (mm yr ⁻¹)	Drainage (mm yr ⁻¹)	Runoff (mm yr ⁻¹)
Conventional	16.756	380.00	887.43	185.26	39.32
Uniform, 25% cut	16.177	285.00	856.41	135.94	25.71
Targeted, 25% cut	16.562	285.00	870.81	125.90	19.86
Uniform, 50% cut	15.029	190.00	806.94	96.34	18.99
Targeted, 50% cut	15.761	190.00	827.70	79.04	14.43
Uniform, 75% cut	13.347	95.00	741.57	68.72	15.83
Targeted, 75% cut	14.018	95.00	752.30	60.01	13.68

442 All per-hectare means use the same 9.866602 million mapped rotation hectares. Dry grain combines wheat and
443 maize harvests; ET includes crops and fallows. Regional grain in million tonnes equals rotation yield multiplied by
444 mapped area in million hectares. Regional water in km³ equals depth in mm multiplied by hectares and 10, divided
445 by 10⁹. The conventional schedule supplies 300 mm to wheat and 80 mm to maize.

446 **3.4. Spatial distribution and persistence of yield contrasts**

447 Relative to uniform allocation, the central and northern latitude bands contributed grain gains of 7.475 and 1.886
448 million t yr⁻¹, while the southern band contributed a loss of 2.142 million t yr⁻¹. The southern band occupied 60.7%
449 of mapped rotation area. Mean rotation-yield contrasts were +2.415, +2.420 and -0.357 t ha⁻¹ yr⁻¹ in the central,
450 northern and southern bands, respectively (Figure 5; Table S9).

451 At the 50% regional irrigation reduction, targeted allocation increased mean annual combined wheat–maize dry
452 grain relative to uniform reduction on 43.92% of the mapped rotation area (4.333 Mha), decreased it on 37.60%
453 (3.710 Mha), and left it unchanged on 18.48% (1.823 Mha). Mean contrasts within the gain and loss areas were
454 +2.439 and -0.903 t ha⁻¹ yr⁻¹, respectively. Their regional contributions were +10.568 and -3.349 Mt yr⁻¹, giving a
455 net regional grain difference of 7.219 Mt yr⁻¹ (0.732 t ha⁻¹ yr⁻¹ across the full mapped area). Local losses offset
456 31.69% of the gross positive contribution (Figure 5d).

457 The area-weighted median grain contrast was 0.000 t ha⁻¹ yr⁻¹, and the 10th and 90th percentiles were -1.182 and
458 +3.553 t ha⁻¹ yr⁻¹. Relative to uniform reduction, mean annual field irrigation increased by 119.0 mm yr⁻¹ in the
459 gain area and decreased by 139.0 mm yr⁻¹ in the loss area. The corresponding crop-plus-fallow ET contrasts were

460 +94.4 and -55.1 mm yr⁻¹. Irrigation changes balanced at regional scale, whereas the area-weighted mean ET
461 contrast was +20.8 mm yr⁻¹.

462 Negative yield contrasts persisted over 2014–2025: 21.34% of the mapped rotation area had lower yields than under
463 uniform allocation in all 12 years, and 37.60% experienced a loss in at least one year. Every response unit with a
464 negative period-mean contrast recorded losses in at least 9 years; all response units with a positive period-mean
465 contrast recorded gains in all 12 years. Across years, the mean annual area exposed to a loss was 35.37%. Mapped
466 cells with negative period-mean contrasts formed 8 connected components under rook adjacency. The largest
467 contained 3.700 Mha of mapped rotation area (99.73% of all loss area), with occupied cell centers extending from
468 32.0 to 36.3°N and 112.8 to 121.5°E.

469 Negative period-mean yield contrasts occupied 32.80%, 37.60%, and 60.08% of mapped rotation area at reductions
470 of 25%, 50%, and 75%, respectively. At the 75% reduction, the area-weighted median contrast became negative
471 (-0.788 t ha⁻¹ yr⁻¹), although gains on 37.04% of the area produced a positive regional balance. Gross positive and
472 negative contributions at that reduction were +14.100 and -7.487 Mt yr⁻¹, leaving a net gain of 6.614 Mt yr⁻¹, below
473 the net gain at the 50% reduction.

474 **3.5. Antecedent water availability and annual irrigation strategies**

475 Four antecedent storage–rainfall classes occurred during 2003–2013. Nine comparison years had eligible GRACE
476 observations, all in the lower-storage category; three years without eligible GRACE observations received
477 conventional irrigation quotas. At the 95% grain target, both storage strata selected an irrigation fraction of 0.75
478 under lower rainfall and 0.50 under higher rainfall. The 98% target selected 1.00 and 0.75, respectively (Table 5).

479 Continuous strategies selected at the 95% and 98% targets retained 96.27% and 99.63% of conventional grain
480 production over the nine years with eligible GRACE observations. Field-irrigation reductions were 130.72 and
481 35.72 mm per rotation; corresponding ET reductions were 36.84 and 6.17 mm (Figure 9; Table S7). The 95%
482 strategy retained at least 95% in 8 of nine years, with a minimum of 92.12%.

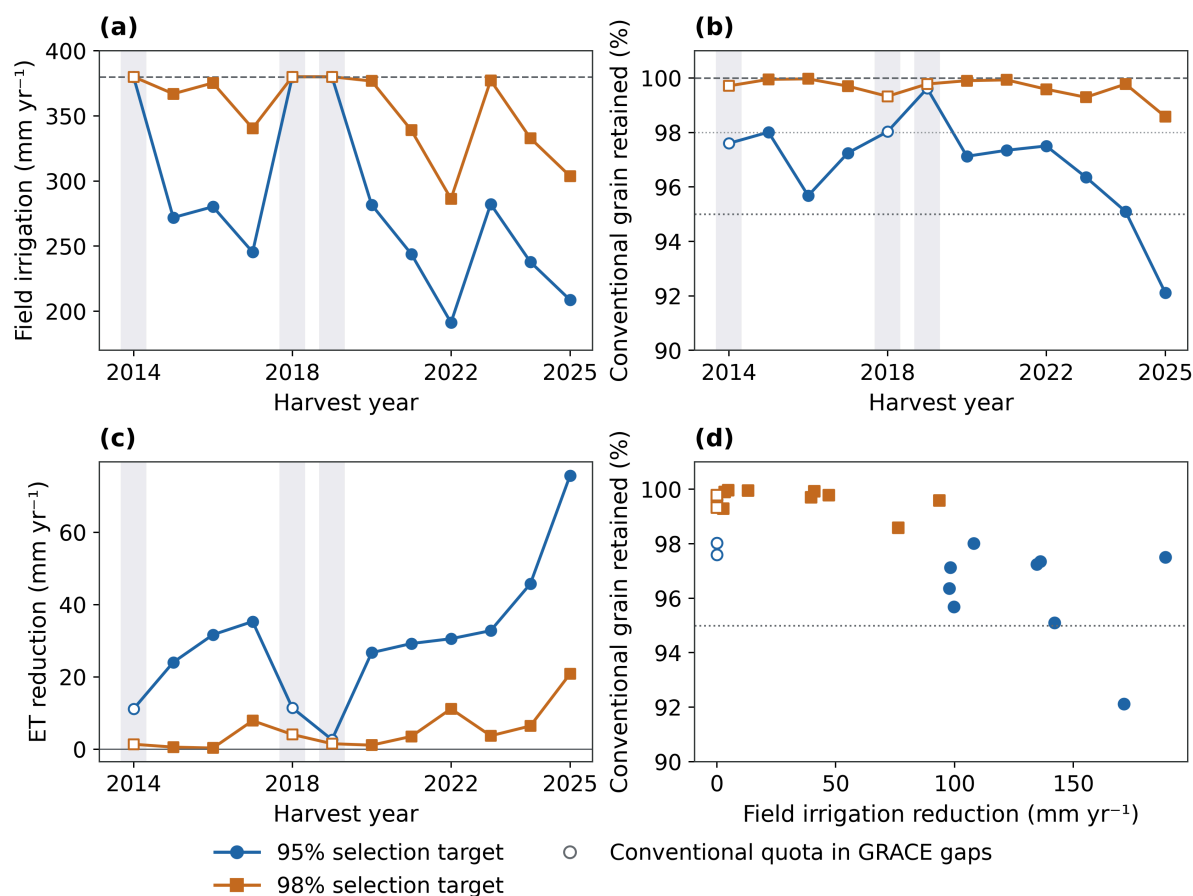


Figure 9. Continuous annual irrigation strategies during 2014–2025. (a) Field irrigation; (b) grain retained relative to conventional irrigation; (c) crop-plus-fallow ET reduction; (d) annual production–irrigation combinations. Blue circles and orange squares identify 95% and 98% selection targets. Shading and open markers identify missing-GRACE years with conventional quotas. Grain and ET can still differ in those years because preceding strategy decisions alter carried soil-water states. Matched rainfall-only histories coincide with the storage–rainfall histories.

Storage–rainfall and matched rainfall-only strategies selected identical irrigation fractions and produced identical grain, ET, drainage and irrigation histories at both targets. Rainfall-only strategies that applied decisions during GRACE gaps followed different irrigation histories and carried different soil-water states into subsequent crops (Table S8).

492 **Table 5. Selected irrigation candidates by relative water-availability class.**

Class	Irrigation fraction	Wheat irrigation (mm)	Maize irrigation (mm)	Selection grain retention (%)
Higher storage dry	0.750	225.000	60.000	95.537
Higher storage wet	0.500	150.000	40.000	97.093
Lower storage dry	0.750	225.000	60.000	96.617
Lower storage wet	0.500	150.000	40.000	97.431

493 Fractions meet the 95% class-mean grain target during selection years 2003–2013. Current shared crop parameters
494 apply throughout. Both storage strata select identical fractions within each rainfall group; higher-storage classes
495 have no later-period exposure.

496 **4. Discussion**

497 We used a calibrated crop model to simulate continuous winter wheat–summer maize rotations across the North
498 China Plain, compare uniform and spatially targeted irrigation under equal water budgets, and test whether
499 antecedent regional water storage improved decisions based on rainfall. Sustaining grain production under restricted
500 irrigation requires consideration of both consumptive water use and the distribution of production losses across
501 locations. The conditional results show that higher regional grain retention can accompany greater
502 evapotranspiration and lower production in some locations, whereas storage information did not change the selected
503 irrigation quotas within the tested design. The management value of these strategies therefore depends on
504 accounting for the complete rotation water balance, soil-water carry-over across crops and fallows, and reliable crop
505 responses to irrigation.

506 **4.1. Marginal irrigation returns and spatial allocation**

507 Spatial targeting exploits differences in marginal grain response to irrigation. Diminishing production gains across
508 field schedules (Yang et al., 2022) and spatial variation in irrigation requirements (Wu et al., 2019) provide a basis
509 for redistributing a limited supply, consistent with regional allocation studies (Jiang et al., 2016). The equal-budget
510 comparison attributes the simulated production difference to delivery allocation. Complete continuous histories
511 retain the effects of earlier irrigation and rainfall on subsequent crop responses.

At the 50% delivery reduction, targeting retained $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ more combined wheat–maize dry grain than uniform cuts. Across 9.8666 million rotation hectares, the corresponding reduction in grain loss was 7.219 million t yr^{-1} , or 4.87% of the uniform total. Targeted production remained below conventional production. The positive contrast in all twelve years indicates consistent regional gains within the simulated record. Its magnitude remains dependent on the crop-response functions and declared management schedules (Table 4; Figure 8).

The spatial distribution sharpens this distinction between regional and local outcomes. Relative to uniform allocation, mean losses occupy 37.60% of mapped rotation area, and losses persist in every comparison year on 21.34%. Gross gains of $10.568 \text{ Mt yr}^{-1}$ are partly offset by losses of 3.349 Mt yr^{-1} ; the area-weighted median is zero despite a positive regional mean. The regional grain objective balances gains and losses among locations. Local production floors constitute an additional allocation constraint.

Regional gain has a pronounced distributional cost. Central and northern band-mean rotation yields increased by 2.415 and $2.420 \text{ t ha}^{-1} \text{ yr}^{-1}$, while the southern mean decreased by $0.357 \text{ t ha}^{-1} \text{ yr}^{-1}$. The southern band covers 60.7% of the rotation area, so its losses substantially offset gains elsewhere (Figure 5; Table S9). The production objective maximizes the area-weighted regional total and supplies no minimum yield for an individual location. Local production floors and constraints on the distribution of irrigation reductions are therefore separate requirements for allocation that protects affected farms.

4.2. Rotation carry-over and the timing of water supply

The spatial production response integrates water effects across both crops. Wheat irrigation can support canopy growth and grain formation, replenish water available to maize or increase drainage. Reduced wheat delivery can increase subsequent dependence on summer rainfall, producing different annual grain responses under different rainfall sequences. The sixteen-year irrigation experiment shows diminishing production gains, and the Wuqiao experiment links precipitation conditions with wheat scheduling and subsequent maize production (Yang et al., 2024; Yang et al., 2022). These coupled responses make the complete rotation the relevant unit for irrigation allocation.

Continuous soil-water histories preserve the delayed benefits and costs of allocation and annual quota changes (Figure S3). Production can be maintained by drawing down profile storage, with later rainfall or irrigation determining its replenishment. A finite-period delivery reduction can therefore include a contribution from stored

539 soil water. Crop-plus-fallow accounting also captures evaporation and drainage between harvest and sowing. Their
540 inclusion connects apparent production retention with the water remaining for the next crop and the complete
541 rotation water balance.

542 The field evidence constrains only part of this carry-over mechanism. Wuqiao simulations use reported total profile
543 storage with an assumed layer distribution. Wheat crop-start storage follows the pre-sowing simulation, whereas
544 maize is independently initialized from treatment-specific observations. Maize irrigation is identical among
545 treatments within each year, and its treatment labels identify preceding wheat schedules (Yang et al., 2024). Maize
546 comparisons thus evaluate crop responses to the initial state associated with wheat management (Figure 7; Figure
547 S2). Regional carry-over additionally depends on the model's continuous propagation and vertical redistribution of
548 water between crops.

549 **4.3. Production, consumptive use and groundwater accounting**

550 The production benefit of targeting carries a consumptive-use trade-off. At the common 50% delivery reduction,
551 targeting increased ET by $2.048 \text{ km}^3 \text{ yr}^{-1}$ and reduced profile drainage by $1.707 \text{ km}^3 \text{ yr}^{-1}$ relative to uniform
552 allocation. With rainfall and total irrigation held constant, differences in ET are balanced by drainage, runoff and
553 soil-storage change (Table 4). A shared delivery constraint therefore permits different consumptive outcomes.
554 Irrigation productivity and consumptive-water savings require distinct objectives (Grafton et al., 2018; Perry et al.,
555 2009).

556 The soil-profile balance also has a different boundary from the aquifer balance. Drainage below 2 m enters a deeper
557 vadose zone, with storage and transit outside the modeled profile (Wu et al., 2023). Reduced field delivery or
558 profile drainage therefore provides insufficient information to quantify changes in pumping, aquifer recharge or
559 groundwater storage. Production and ET comparisons can identify agronomic trade-offs, while groundwater
560 outcomes require the corresponding subsurface and abstraction accounting.

561 GRACE offers regional storage context at a broader scale. Its anomalies integrate groundwater with other terrestrial
562 stores, and its effective observation scale exceeds the displayed grid resolution (Save et al., 2016; Scanlon et al.,
563 2016). Recent aquifer recovery reflects water transfers, abstraction management and hydroclimatic conditions
564 (Long et al., 2025; Xu et al., 2026). The crop-model scenarios quantify changes in field delivery, consumption and
565 profile outflow without attributing those observed storage changes to individual irrigation strategies.

566 **4.4. When storage information changes an irrigation decision**

567 A storage indicator adds decision value when its inclusion changes the selected action or improves the grain–water
568 outcome under common constraints. GRACE represents regional storage conditions, while antecedent rainfall
569 describes more local water input (Save et al., 2016; Scanlon et al., 2016). Their different physical support motivates
570 joint classification, but decision value also depends on how the resulting classes distinguish among the available
571 irrigation fractions.

572 Both storage strata selected the same fraction for a given rainfall group at the 95% and 98% grain targets. Storage–
573 rainfall and matched rainfall-only strategies consequently produced identical complete histories of irrigation, crop
574 states, grain, ET and drainage. The storage split supplied additional environmental description without changing an
575 irrigation action in this five-level quota design. All nine later years with eligible observations fell in the lower-
576 storage category, so the higher-storage choices also lacked later-period exposure (Table 5; Figure 9). These features
577 delimit the configuration and conditions represented by the equal outcomes.

578 Monitoring coverage must remain distinct from the grouping criterion. Rainfall-available controls can change
579 irrigation during GRACE gaps, and those earlier decisions alter the soil-water state inherited in later years. Their
580 outcomes therefore combine rainfall-based selection with additional years of management. Matched coverage
581 preserves the appropriate comparison for the incremental effect of storage information. Finer quota choices,
582 alternative thresholds and verified release availability would define different decision experiments with their own
583 selection and evaluation requirements.

584 **4.5. Crop-response constraints and management reliability**

585 Management selection depends on the modeled marginal response to water, in addition to absolute grain and canopy
586 agreement. Intercepted-PAR radiation-use efficiency (Kiniry et al., 1989), cohort-specific leaf area and thermal
587 senescence connect canopy development with carbon supply; grain number and fixed individual grain mass
588 constrain the sink. Root uptake and the assimilation stress exponent then determine how water shortage changes
589 production. Variation among wheat-model transpiration formulations demonstrates the sensitivity of water-use
590 predictions to these processes (Cammarano et al., 2016). Water coefficients near their prescribed bounds remain
591 effective empirical estimates whose physiological interpretation and uniqueness are limited.

592 The estimation and evaluation data provide different evidence for these responses. Earlier growth estimates use 480
593 wheat and 433 maize observations across four wheat sites and five maize sites, with whole site-years and balanced
594 site contributions. Incomplete station irrigation histories limit transfer comparisons, and station observations have

no weight in the fitted water-response objective. Wuqiao 2016–2018 supplies the documented treatment seasons for that fit. Wuqiao 2019 supplies a single test year with reported total storage, assumed layer distributions and transferred soil hydraulics; prior inspection of its outcomes makes the test retrospective (Figures 3, 6 and 7). Regional comparison years also overlap crop-fitting years, so regional production differences characterize model scenarios rather than time-forward crop validation.

The absolute regional production reference also depends on the prescribed management and crop parameters. Conventional simulations produce 8.065 t ha⁻¹ of wheat dry grain and 8.691 t ha⁻¹ of maize dry grain per year. These values represent fixed irrigation dates, crop calendars and nutrition modifiers rather than an independently validated regional harvest inventory. The 0.732 t ha⁻¹ yr⁻¹ allocation advantage depends on the same simulated marginal responses; its regional scaling does not remove field-response uncertainty. Substantial field-test errors constrain confidence in irrigation-response transfer. Wheat grain nRMSE was 83.27%, with a bias of +3.108 t ha⁻¹ at the published 13% moisture basis; the absolute W3–W0 ET-contrast error was 116.37 mm. Maize ET NSE was negative, with an ET-contrast error of 52.74 mm (Table 3; Table S5). These discrepancies affect the response functions used to select regional allocations. Numerical water-balance closure demonstrates conservation but supplies no measure of response accuracy. The eight purposive actual-cell checks address a separate uncertainty, spatial aggregation, and provide no regional field-validation sample. Depth-resolved soil water and complete dated management would better constrain uptake and redistribution; canopy and soil-moisture assimilation also requires treatment of retrieval errors and observation operators (Ines et al., 2013).

Production-retention targets introduce a further management distinction between period means and annual reliability. Mean retention of 96.27% and 99.63% coincided with annual minima of 92.12% and 98.58%; the 95% strategy met that threshold in eight of nine class-available years (Figure 9). A mean target consequently supplies no shortfall probability or guarantee for an individual farm. Local production floors, acceptable shortfall frequency and a reserve entering the next crop can complement regional delivery and consumption constraints. Forecast-based irrigation studies illustrate explicit treatment of water-stress risk (Roy et al., 2021). Field scheduling additionally requires current soil-water information, weather forecasts and application timing.

4.6. Limitations and future directions

The principal limitation is uncertainty in the crop responses used for strategy selection. Field testing covers a single, previously inspected year, while incomplete station management and assumed initial water profiles constrain transfer. Remaining errors in ET and grain responses prevent the simulated policy advantages from being

624 interpreted as independently validated field benefits. Further evaluation requires new site-years with complete
625 management records, withheld from both model development and strategy selection. Concurrent measurements of
626 soil water by depth, ET, canopy development and harvest would allow separate assessment of water redistribution,
627 crop stress and irrigation-treatment contrasts.

628 Regional results also depend on a static rotation map, representative climate–soil units, prescribed irrigation dates
629 and fixed nutrition modifiers. Checks at purposively selected source cells cannot bound regional uncertainty.

630 Dynamic nitrogen cycling is absent, and changes in crop area or actual farm practices are not reconstructed. Future
631 analyses should expand evaluation at individual source cells and propagate uncertainty in crop parameters, soil
632 hydraulics, weather and spatial aggregation through paired policy histories. Allocation stability and the magnitude
633 and persistence of local losses require uncertainty assessment alongside the regional mean.

634 The storage comparison is specific to the tested quota grid, classification thresholds and lower-storage conditions
635 represented during later comparison years. GRACE release availability before sowing remains unverified, so the
636 scenarios do not establish operational scheduling performance. Prospective decision experiments should use
637 information available on each decision date, maintain matched rainfall-only controls and evaluate both storage
638 states. Finer quotas and application timing informed by soil water could be tested with constraints on consumptive
639 use, local production and the risk of production shortfalls.

640 Finally, the profile uses free drainage at its lower boundary, with no capillary supply or deeper subsurface storage,
641 and field irrigation is not linked to measured abstraction. Groundwater applications require joint accounting of soil
642 water and aquifer flows, constrained by observations of pumping, recharge and storage change. Delivery capacity,
643 irrigation costs and farmer participation also require assessment before the agronomic scenarios can support
644 operational management.

645 **5. Conclusions**

646 At an identical 50% irrigation reduction, targeted allocation retained $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ more simulated combined
647 wheat–maize dry grain than uniform cuts during 2014–2025. This reduced regional grain loss by 7.219 million t
648 yr^{-1} (4.87% of uniform production), while increasing crop-plus-fallow ET by 20.76 mm yr^{-1} ($2.048 \text{ km}^3 \text{ yr}^{-1}$).

649 Relative to uniform allocation, mean grain losses occurred on 37.60% of mapped rotation area and persisted in all
650 twelve years on 21.34%. Regional production retention therefore requires separate consumption and local
651 production constraints.

652 Annual strategies selected at 95% and 98% grain-retention targets retained 96.27% and 99.63% of conventional
653 production over nine years with eligible GRACE observations. The 95% strategy reduced irrigation by 130.72 mm
654 and ET by 36.84 mm per rotation. Matched rainfall-only controls selected identical quotas and produced identical
655 outcomes, so storage classification provided no additional decision benefit within the tested quota design and
656 monitoring coverage.

657 Continuous rotation accounting preserves the influence of preceding management on subsequent production and
658 water use. The field-test errors and incomplete station management delimit confidence in transferring the estimated
659 crop responses. Translation of these regional scenarios into irrigation management depends on observed response
660 accuracy, explicit production-risk constraints and field scheduling informed by soil water and forecasts.
661 Groundwater effects additionally require pumping, recharge and deeper subsurface storage beyond the crop-model
662 boundary.

663 **Data and code availability**

664 Model code, frozen calibration and regional inputs, observation subsets and quality exclusions, weather, calibrated
665 parameters, predictions, evaluation metrics and figure-data tables are available at [https://github.com/SmartAG-](https://github.com/SmartAG-Team/agri_water_management)
666 [Team/agri_water_management](https://github.com/SmartAG-Team/agri_water_management). The repository includes a standalone Open Crop Model package, pinned
667 dependencies, reproduction commands and SHA-256 provenance. Reproduction uses the included inputs without
668 another model checkout or access to the historical workspace. Original datasets retain their provider licences and
669 attribution requirements, documented in the repository source catalogue.

670 **Declaration of generative AI and AI-assisted technologies**

671 AI assisted with grammar checking.

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